디지털 논리

Counters (1)

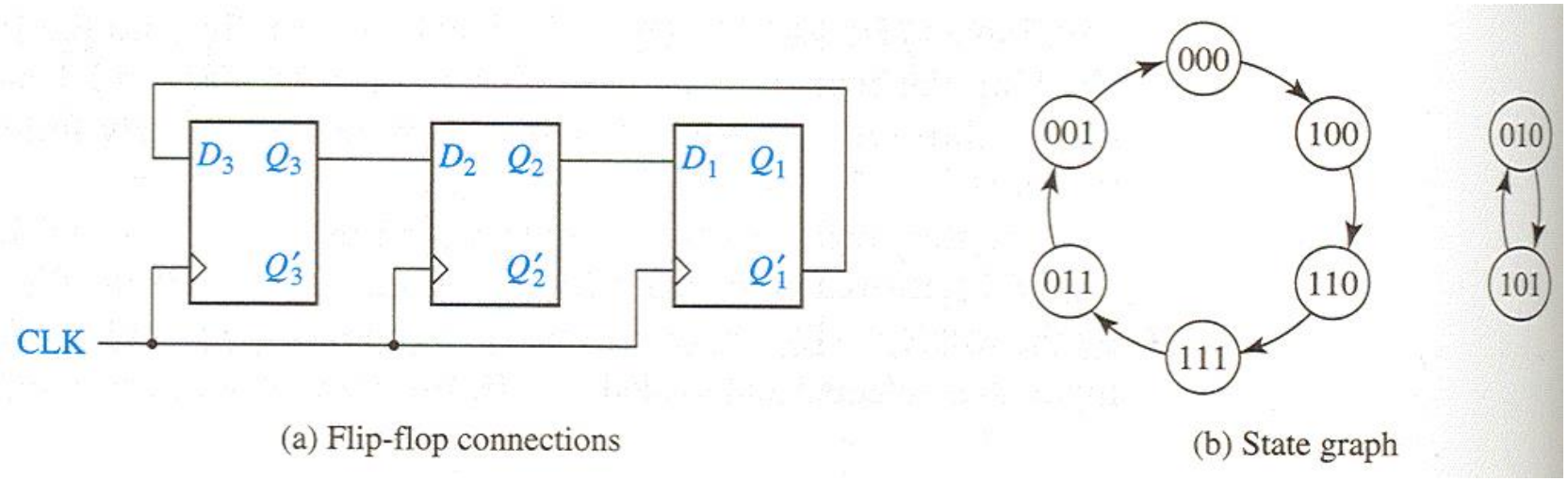
이 상 화

Counters

- Counters
  - Changing states (n-bit pattern) in a predefined sequence at the clock pulse (rising/falling edge)
  - Consisting of n flip-flops(FF) with a common clock
  - No inputs
  - Feedback of FF outputs

Shift Register with Feedback

- Shift register with negative feedback
- Successive states transition (state graph)
- A circuit that cycles through a fixed sequence of states is called *counter* .
  - Johnson counter to have negative feedback



(a) Flip-flop connections (b) State graph

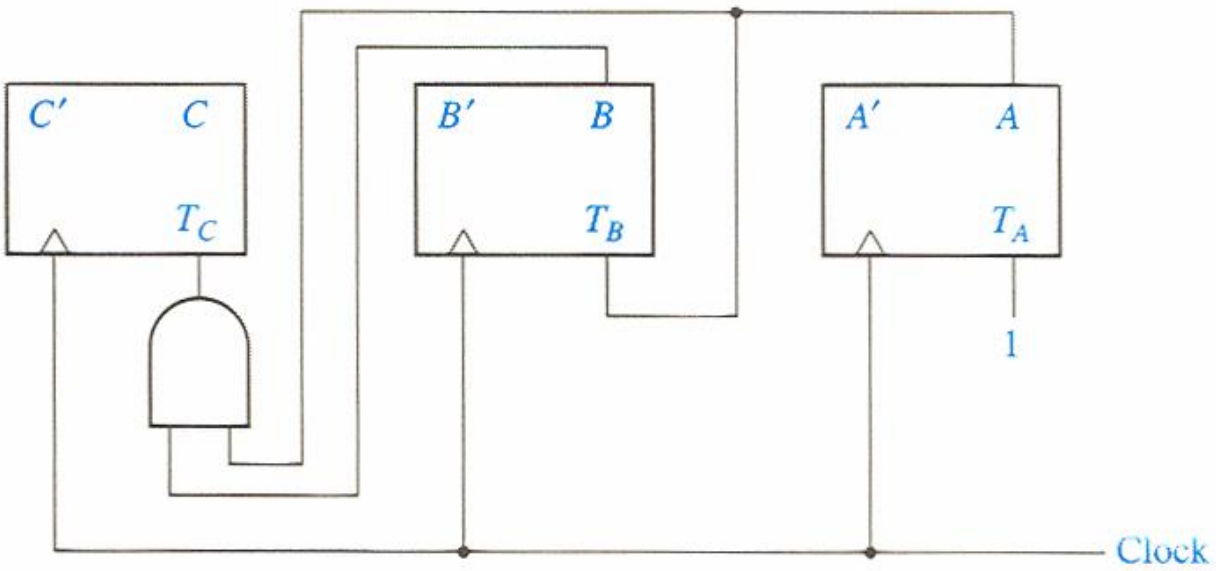
When initial state $D_3D_2D_1$=000,

there are no states 010 and 101

Ex: Binary Counter Using T-FFs

- **3 bit binary counter:**

=> 000 -> 001 -> 010 -> 011 -> 100 -> 101 -> 110 -> 111 -> 000



Counter using 3 T-FFs (8 counts)

How to design the counters using specific FFs ??

Counter Design

- How to design the counters using the specific FFs?

1) Define the state table for present and next states using the predefined sequence

2) Determine the input of each FF for the state transition

Design the inputs of FFs using the outputs (present states) of FFs

Binary Counters Using T-FFs (1)

State table for state sequence

Present State			Next State		
C	B	A	C^+	B^+	A^+
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

Inputs of FFs for state change

Flip-Flop Inputs		
T_C	T_B	T_A
0	0	1
0	1	1
0	0	1
1	1	1
0	0	1
0	1	1
0	0	1
1	1	1

Binary Counters Using T-FFs (2)

Karnaugh map for FF inputs

BA \ C	0	1
00	0	0
01	0	0
11	1	1
10	0	0

T_C

BA \ C	0	1
00	0	0
01	1	1
11	1	1
10	0	0

T_B

Present State			Next State			Flip-Flop Inputs		
C	B	A	C^+	B^+	A^+	T_C	T_B	T_A
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

$T_C = BA$

$T_B = A$

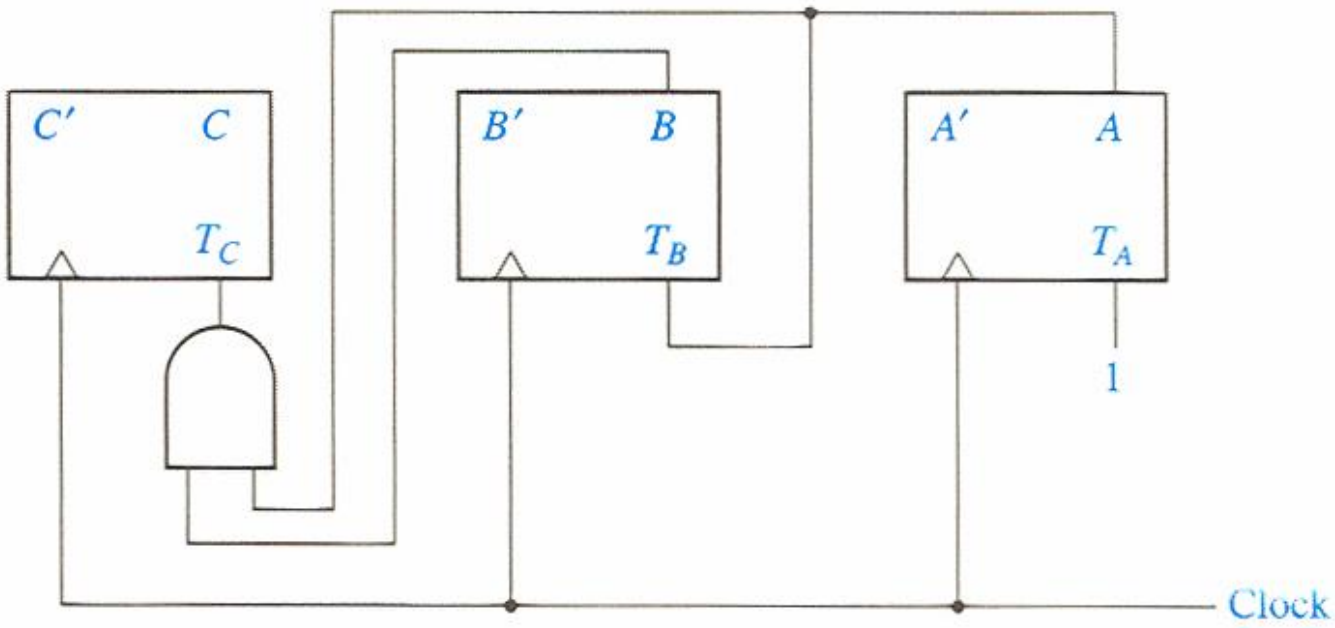
$T_A = 1$

FF inputs are determined with Q and Q+

Binary Counter Using T-FFs (3)

- **3 bit binary counter:**

=> 000 -> 001 -> 010 -> 011 -> 100 -> 101 -> 110 -> 111 -> 000



$T_C = BA$ $T_B=A$ $T_A=1$

Binary Counter Using D-FFs (1)

State Table

Present State			Next State		
C	B	A	C^+	B^+	A^+
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

Inputs of D−FFs

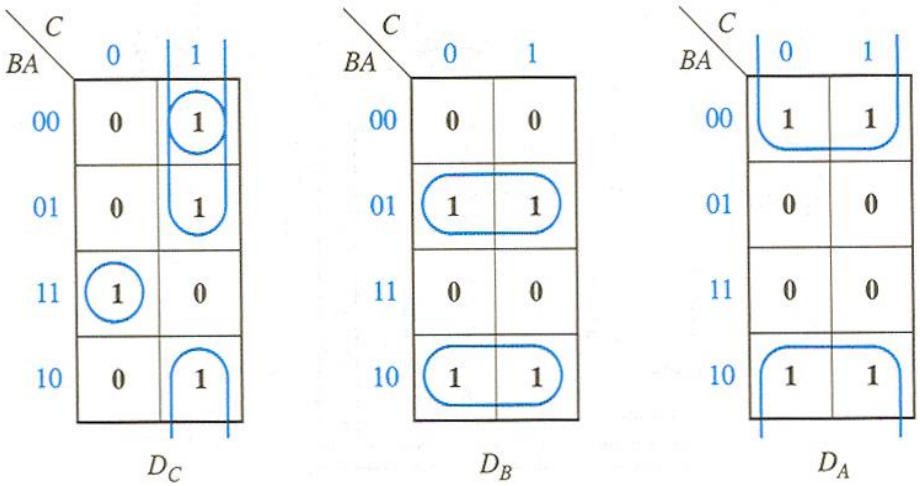
D_C	D_B	D_A
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1
0	0	0

Binary Counter Using D-FFs (2)

State Table

Karnaugh map for inputs of D−FFs

Present State C	B	A	D_C	D_B	D_A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0



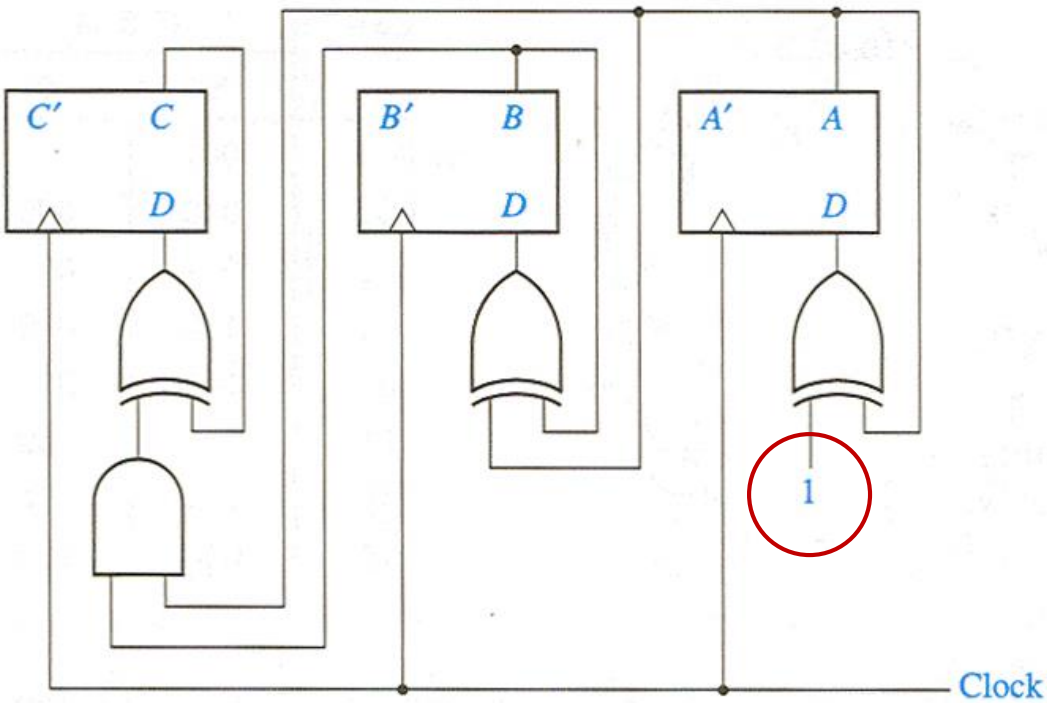
$$D_A = A^+ = A'$$

$$D_B = B^+ = BA' + B'A = B \oplus A$$

$$D_C = C^+ = C'BA + CB' + CA' = C'BA + C(BA)' = C \oplus BA$$

Binary Counter Using D-FFs (3)

- Using D Flip-Flops and XOR gates



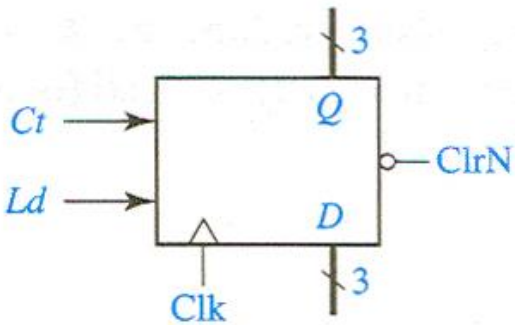
$$D_A = A^+ = A'$$

$$D_B = B^+ = BA' + B'A = B \oplus A$$

$$D_C = C^+ = C'BA + CB' + CA' = C'BA + C(BA)' = C \oplus BA$$

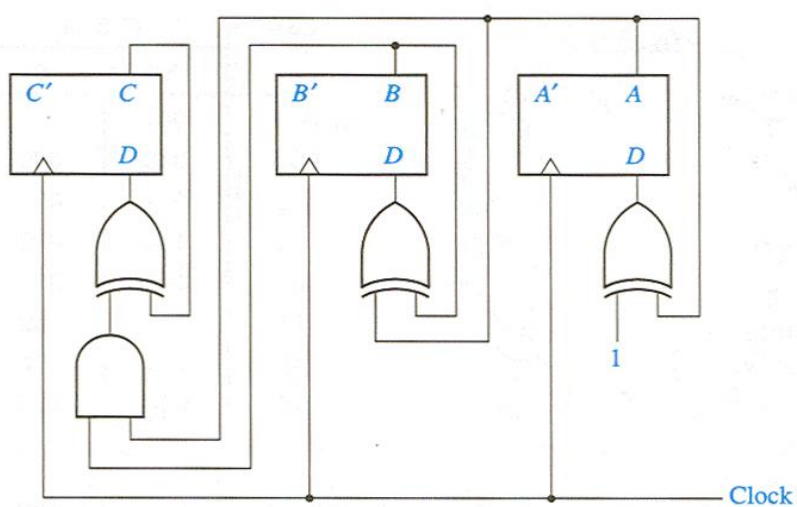
Binary Loadable Counter (1)

- Loadable Counter with counter enable: Symbol and state table



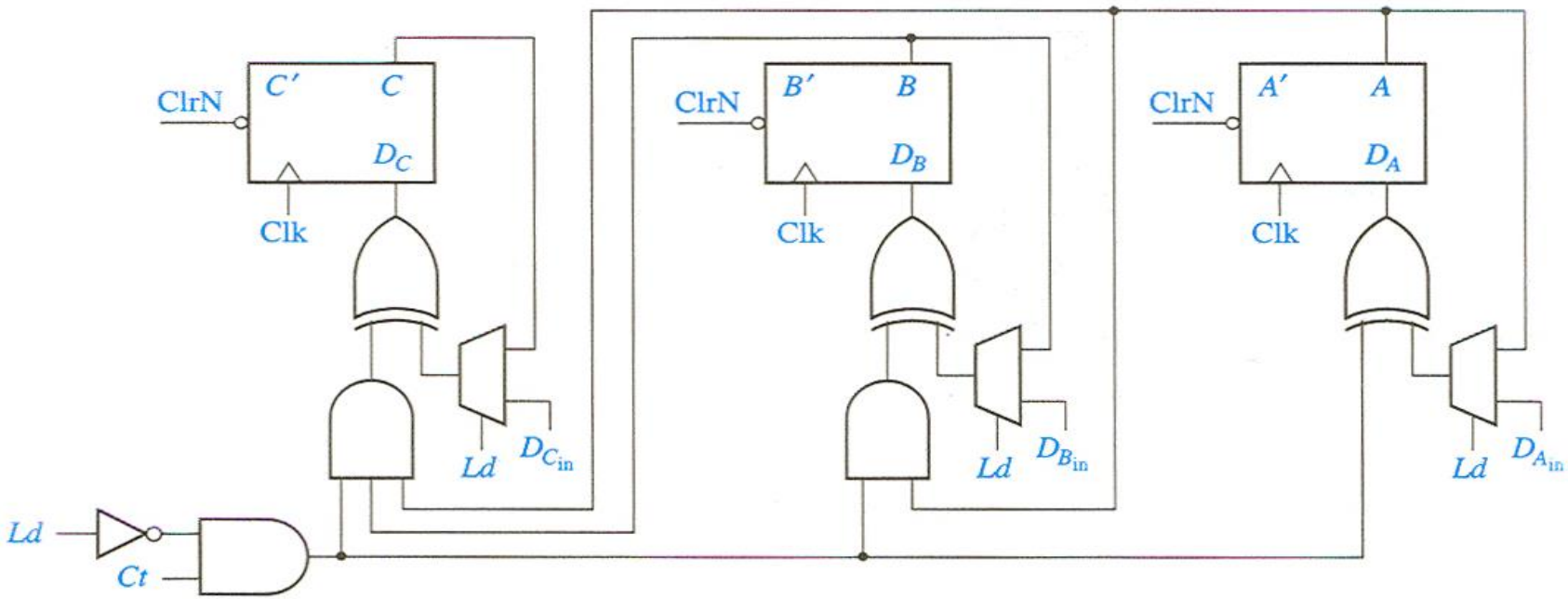
ClrN	Ld	Ct	C^+	B^+	A^+	
0	X	X	0	0	0	
1	1	X	D_C	D_B	D_A	(load)
1	0	0	C	B	A	(no change)
1	0	1	Present state + 1			

- Using the binary counter with D-FFs and XOR gates



Binary Loadable Counter (2)

- Logic circuit using Multiplexer and XOR gates



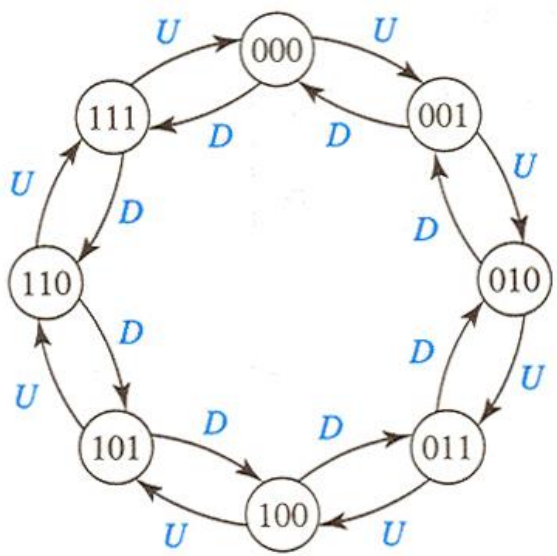
$$A^+ = D_A = (Ld' \cdot A + Ld \cdot D_{Ain}) \oplus Ld' \cdot Ct$$

$$B^+ = D_B = (Ld' \cdot B + Ld \cdot D_{Bin}) \oplus Ld' \cdot Ct \cdot A$$

$$C^+ = D_C = (Ld' \cdot C + Ld \cdot D_{Cin}) \oplus Ld' \cdot Ct \cdot B \cdot A$$

Up-Down Binary Counter (1)

- State graph and state table for Down and Up variables



CBA	$C^+B^+A^+$	
	U	D
000	001	111
001	010	000
010	011	001
011	100	010
100	101	011
101	110	100
110	111	101
111	000	110

Down (D)

D_B	D_C
1	1
0	0
0	0
1	0
1	0
0	1
0	1
1	1

- Logic equation

$$D_A = A^+ = A \oplus (U + D)$$
$$D_B = B^+ = B \oplus (UA + DA')$$
$$D_C = C^+ = C \oplus (UBA + DB'A')$$

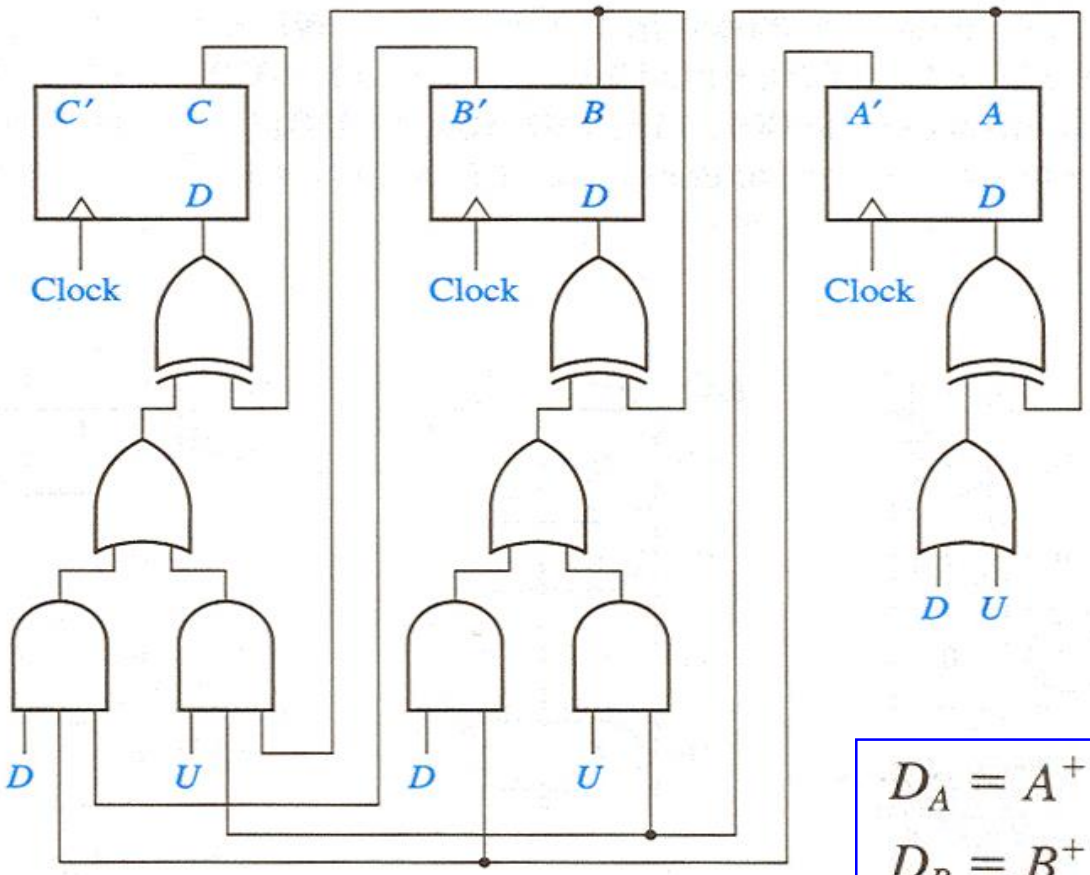
$D_A = A'$

$D_B = B'A' + BA = B \oplus A'$

$D_c = C'B'A' + CA + CB$
$= C'B'A' + C(B+A)$
$= C'B'A' + C(B'A')'$
$= C \oplus B'A'$

Up-Down Binary Counter (2)

- Logic circuit



$$D_A = A^+ = A \oplus (U + D)$$

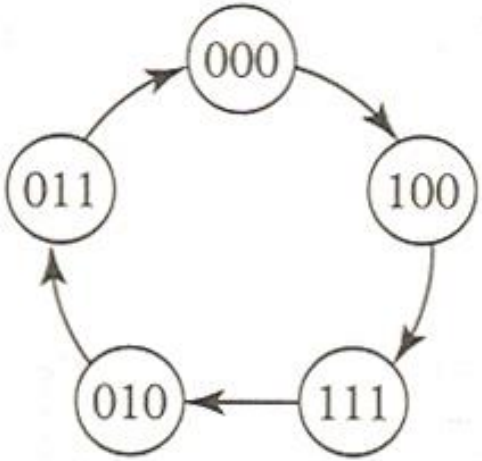
$$D_B = B^+ = B \oplus (UA + DA')$$

$$D_C = C^+ = C \oplus (UBA + DB'A')$$

Other Sequences Counter (1)

- Sequence is not in straight binary order
- Define state graph

- Design the counter using FFs
  - Form a state table with present and next states
  - Find next state k-map and inputs K-map
  - Find the logic equation of FF inputs and draw the circuit

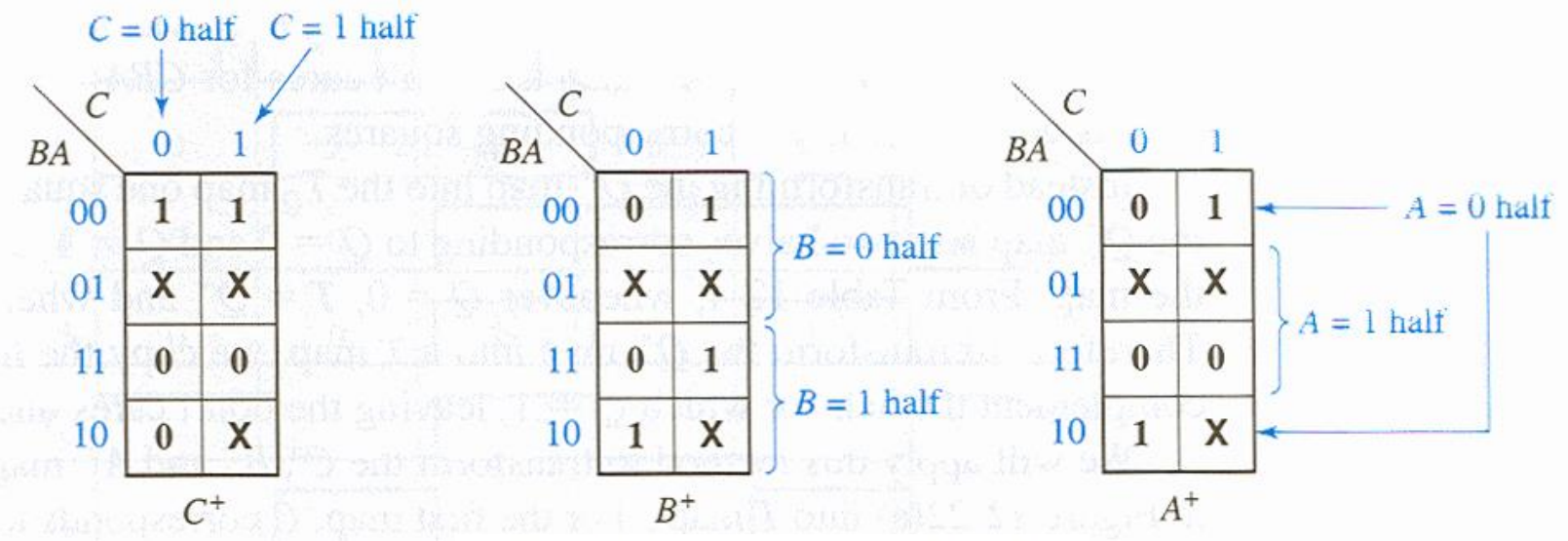


C	B	A	C^+	B^+	A^+
0	0	0	1	0	0
0	0	1	–	–	–
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	1	1
1	0	1	–	–	–
1	1	0	–	–	–
1	1	1	0	1	0

An example of predefined sequence and state table

Other Sequences Counter (2)

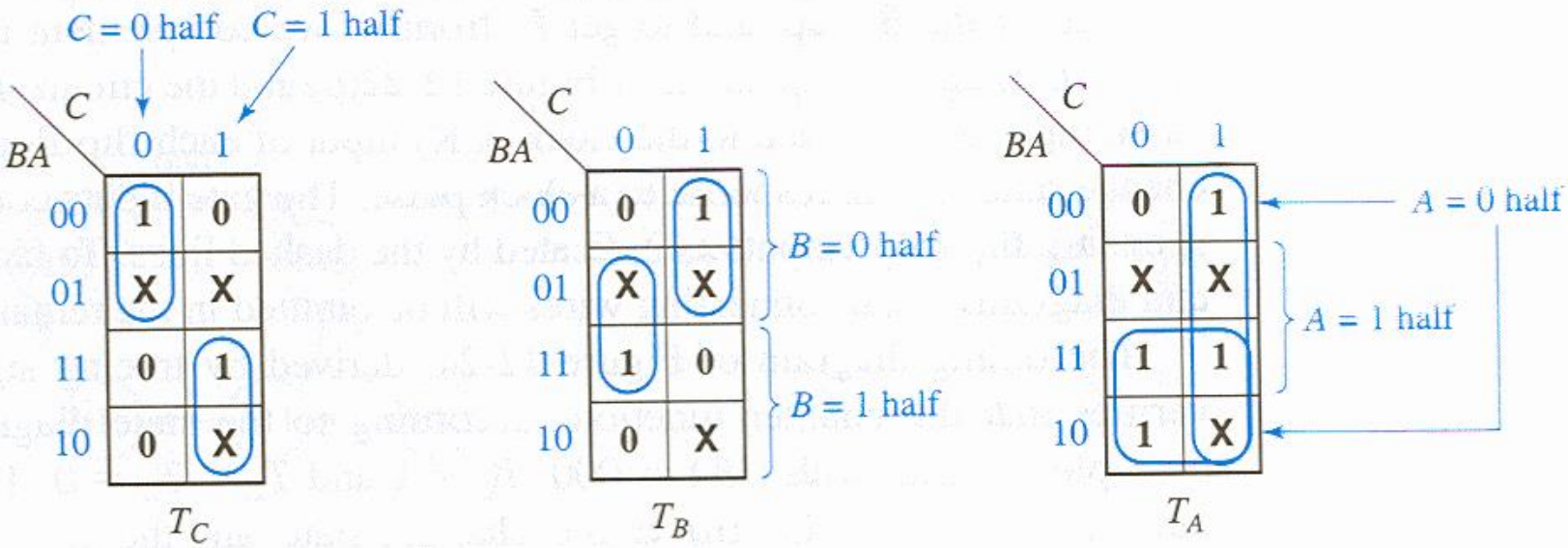
- State map for the next states

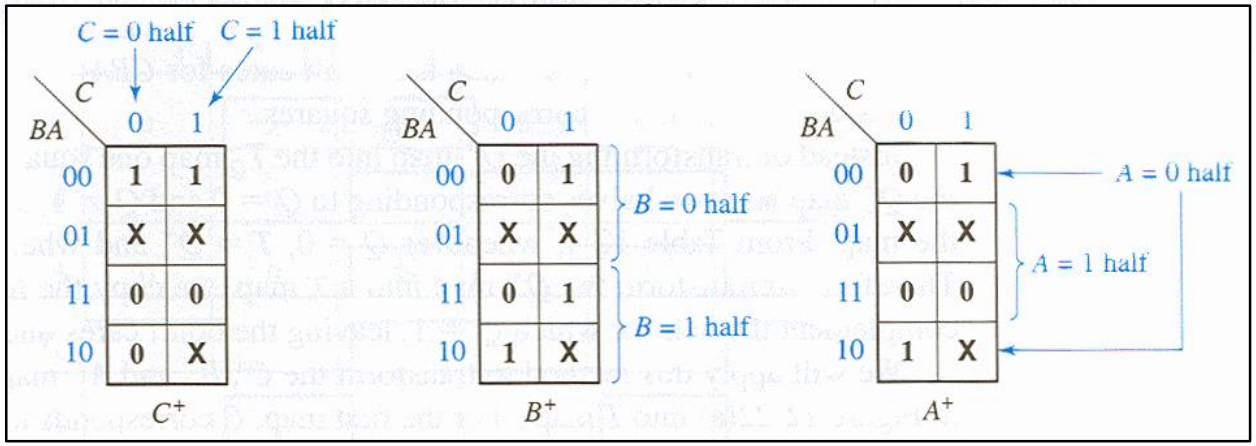


C	B	A	C^+	B^+	A^+
0	0	0	1	0	0
0	0	1	–	–	–
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	1	1
1	0	1	–	–	–
1	1	0	–	–	–
1	1	1	0	1	0

Other Sequences Counter (3)

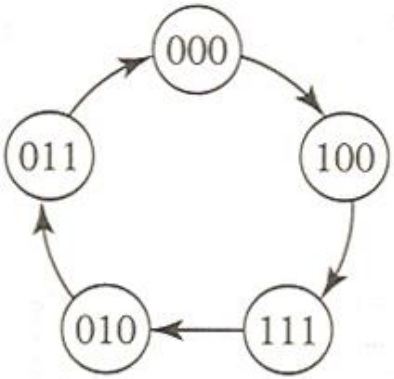
- Karnaugh map for the inputs of T−FFs

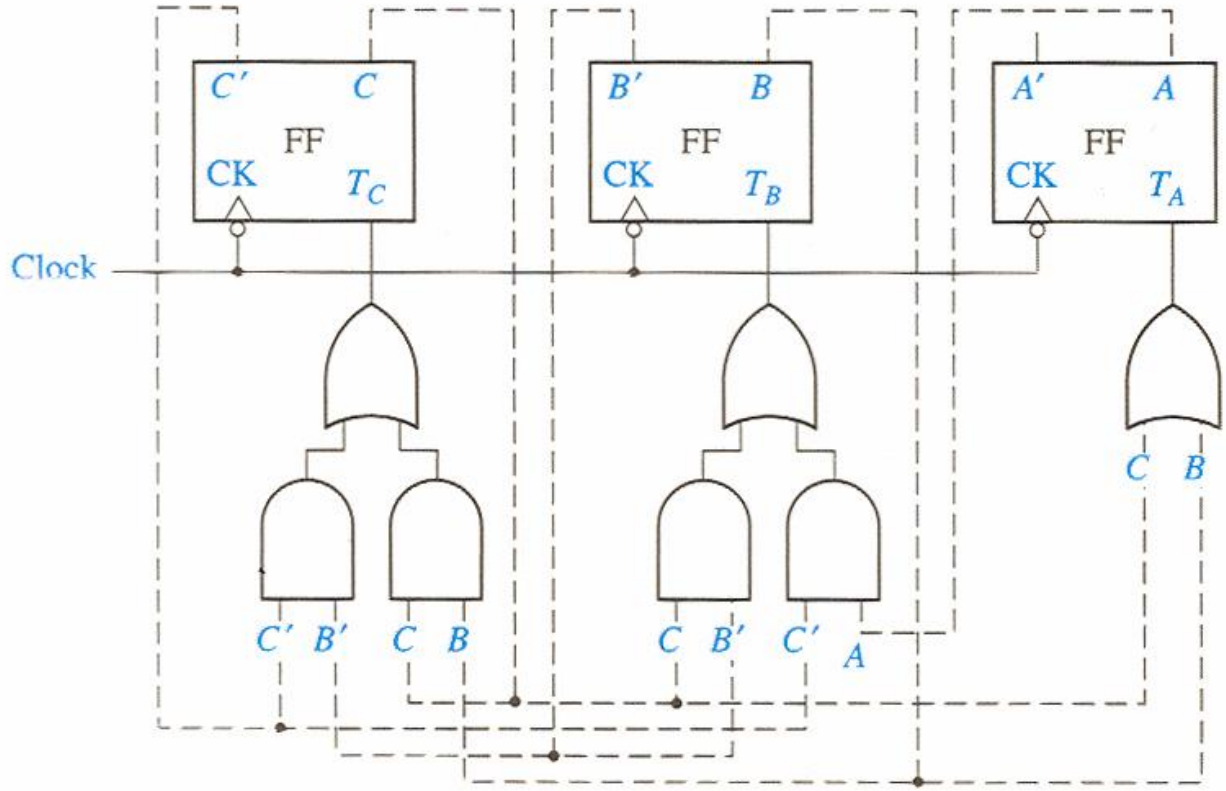




Other Sequences Counter (4)

- Circuit using T-FFs

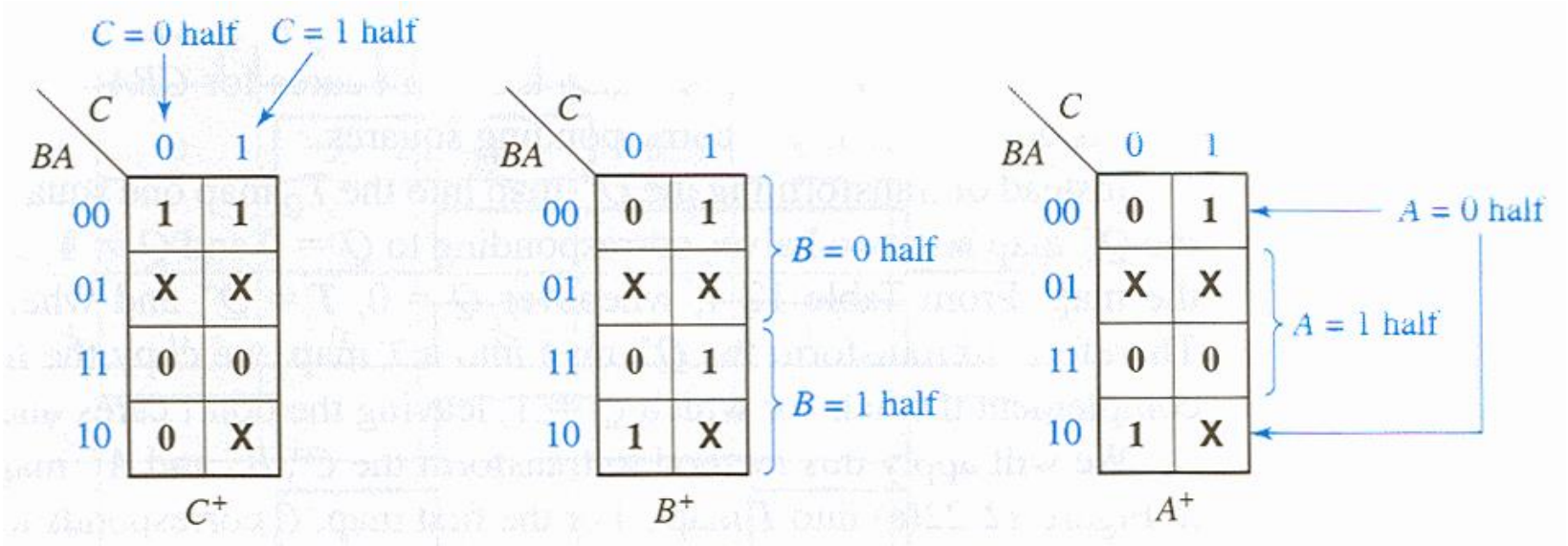




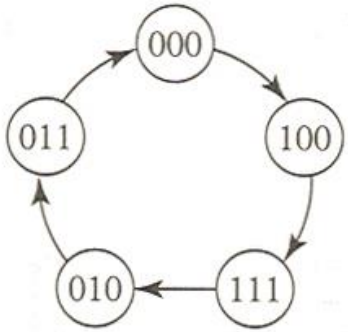
Other Sequences Counter (5)

- Using D-FFs: D (input of D-FF) = Q^+ (next state)

  => input map is the same with next state map



D_C D_B D_A

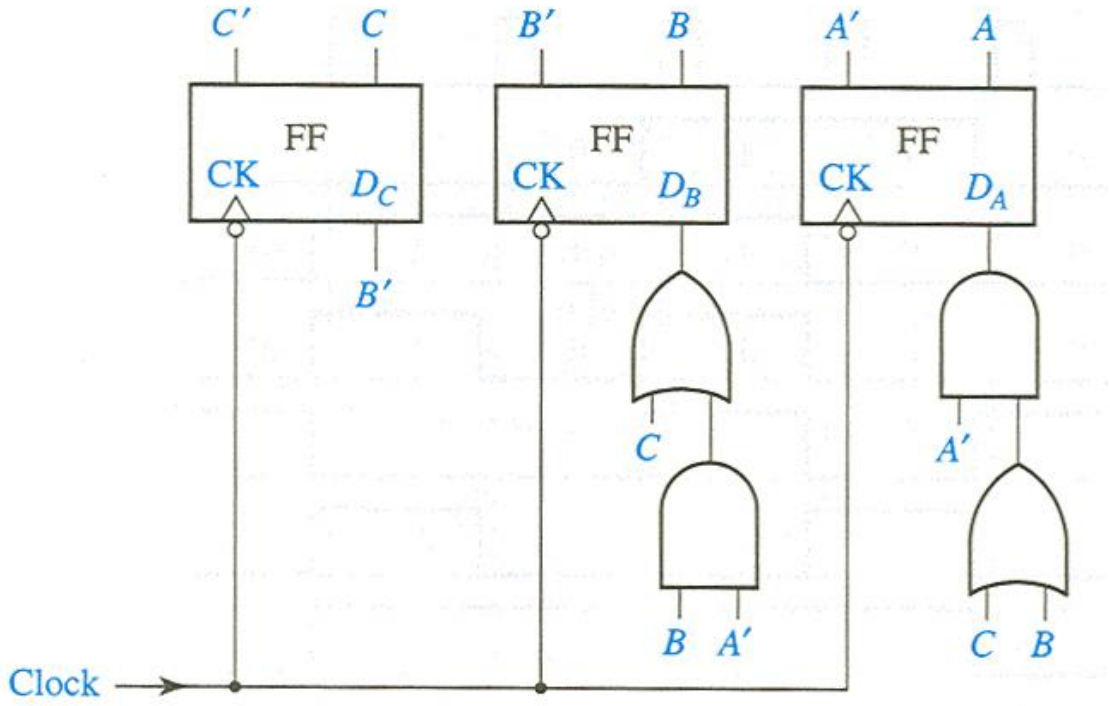


$$D_C = C^+ = B' \qquad D_B = B^+ = C + BA'$$

$$D_A = A^+ = CA' + BA' = A'(C + B)$$

Other Sequences Counter (6)

- Counter circuit using D Flip-Flops

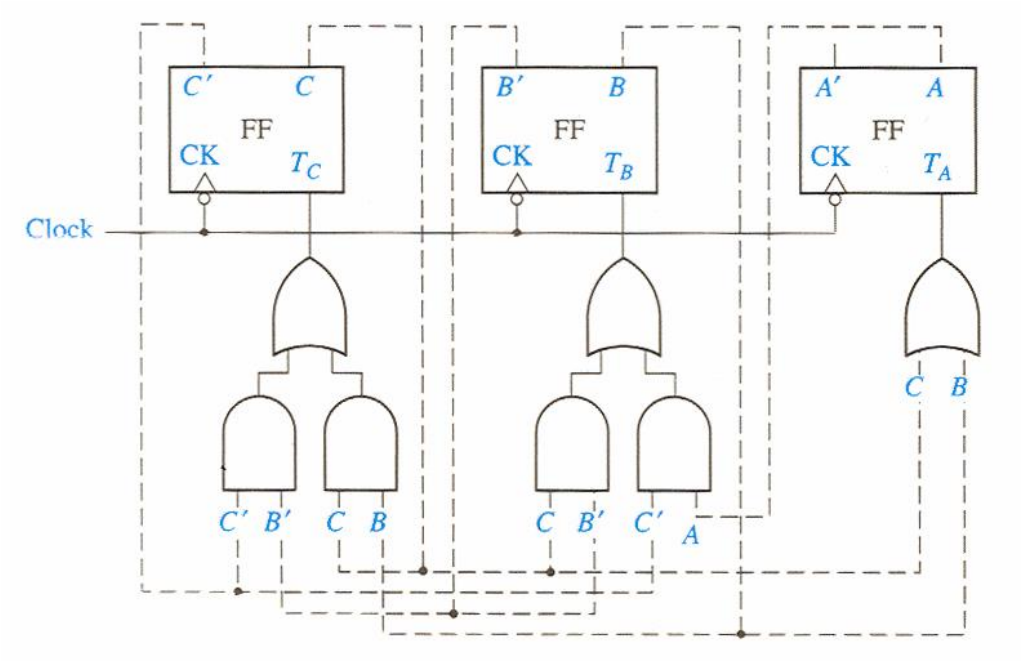


$$D_C = C^+ = B' \qquad D_B = B^+ = C + BA'$$

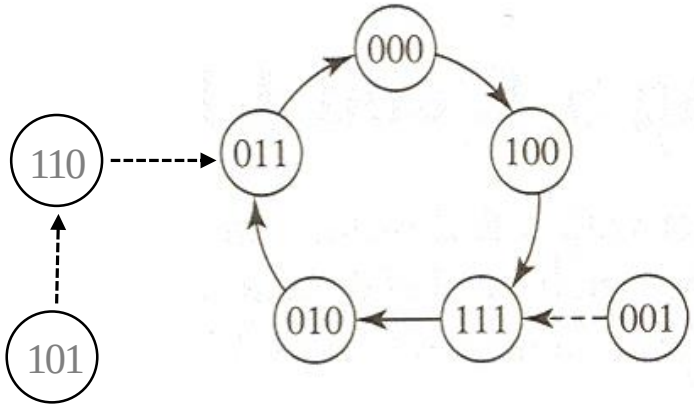
$$D_A = A^+ = CA' + BA' = A'(C + B)$$

Complete Sequences of Counter (1)

- Complete state graph of T−FF



CBA	$T_C T_B T_A$	$C^+B^+A^+$
001	1 1 0	1 1 1
101	0 1 1	1 1 0
110	1 0 1	0 1 1

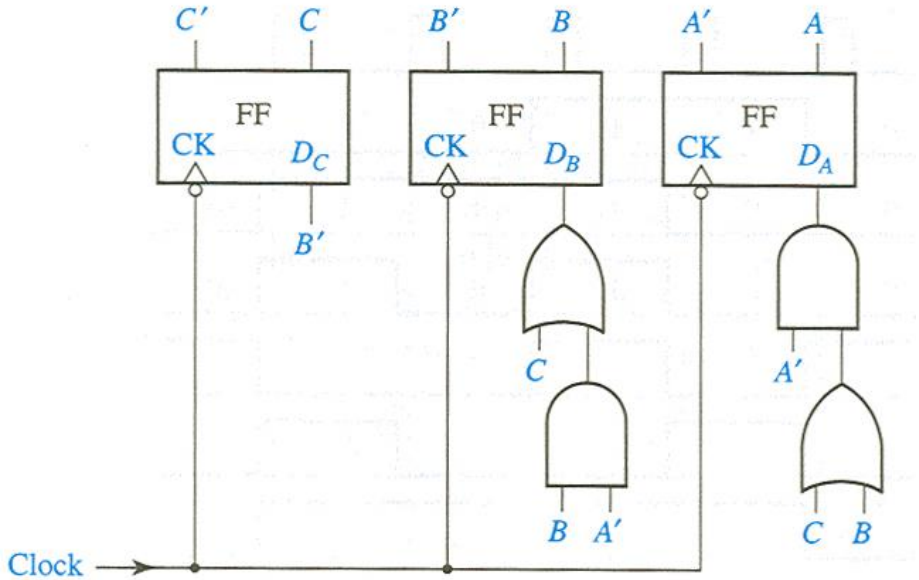


$T_C = C'B' + CB$ $T_B = C'A + CB'$ $T_A = C + B$

Complete state graph

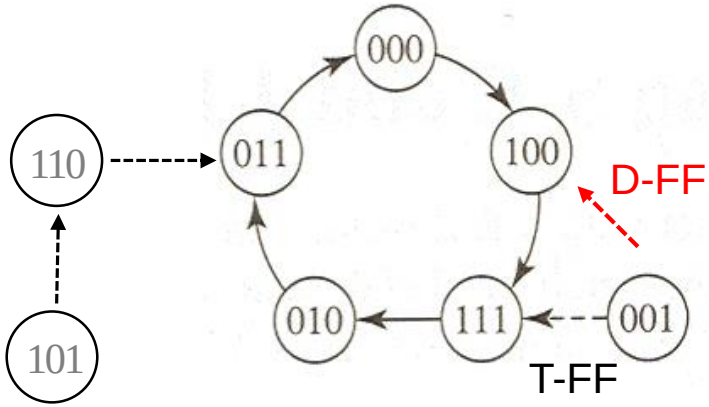
Complete Sequences of Counter (2)

- Complete state graph of D−FF



CBA	$D_C D_B D_A$	$C^+B^+A^+$
001	1 0 0	1 0 0
101	1 1 0	1 1 0
110	0 1 1	0 1 1

$$D_C = C^+ = B' \qquad D_B = B^+ = C + BA'$$
$$D_A = A^+ = CA' + BA' = A'(C + B)$$



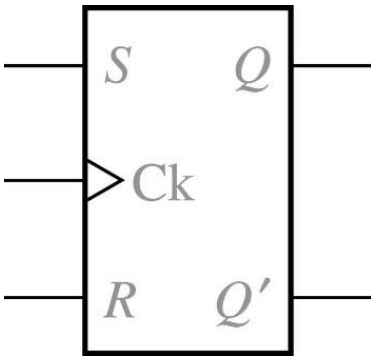
Complete state graph

디지털 논리

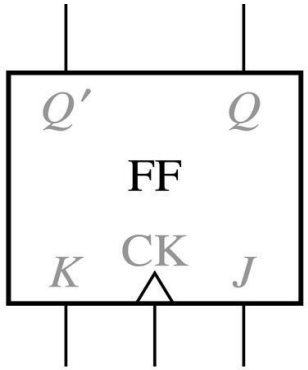
Counter (2)

이 상 화

Review of S-R, J-K FF



S=R=0	No state change
S=1, R=0	Set Q to 1 (after active Ck edge)
S=0, R=1	Reset Q to 0 (after active Ck edge)
S=R=1	Not allowed



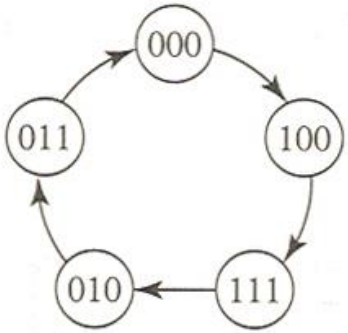
J=K=0	No state change
J=1, K=0	Set Q to 1 (after active Ck edge)
J=0, K=1	Reset Q to 0 (after active Ck edge)
J=K=1	Toggled

Counter Design with S-R FFs (I)

1) Find next state table and map from state graph
2) Find state table for S−R inputs using the next state map
3) Find logic equation of S−R inputs and draw the circuit

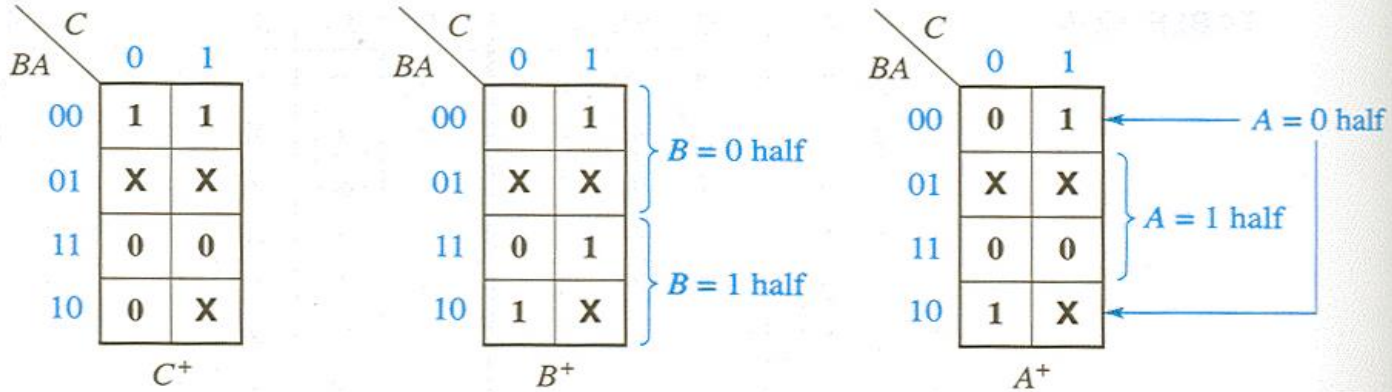
Next state table S−R inputs table

C	B	A	C^+	B^+	A^+	S_C	R_C	S_B	R_B	S_A	R_A
0	0	0	1	0	0	1	0	0	X	0	X
0	0	1	–	–	–	X	X	X	X	X	X
0	1	0	0	1	1	0	X	X	0	1	0
0	1	1	0	0	0	0	X	0	1	0	1
1	0	0	1	1	1	X	0	1	0	1	0
1	0	1	–	–	–	X	X	X	X	X	X
1	1	0	–	–	–	X	X	X	X	X	X
1	1	1	0	1	0	0	1	X	0	0	1



Counter Design with S-R FFs (II)

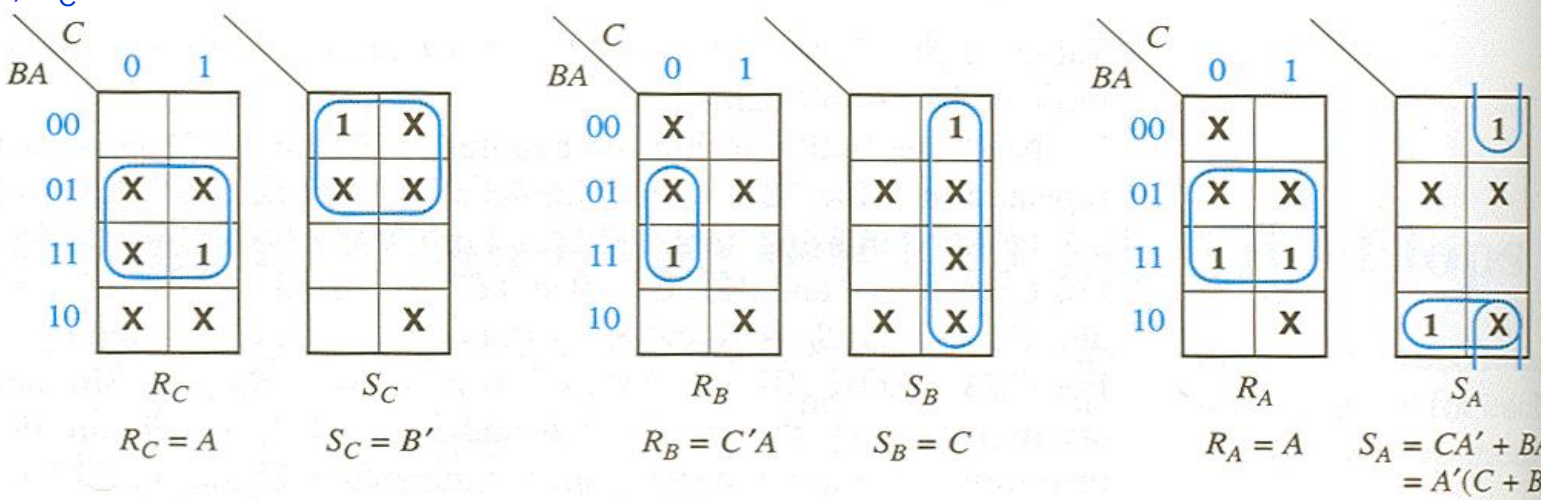
Find maps of S−R inputs from the next state maps and S−R FF operation



(a) Next-state maps

C=0 일 때, C^+=1 되려면

S_C=1, R_C=0 이어야 한다.



(b) S-R flip-flop equations

Counter Design with S-R FFs (III)

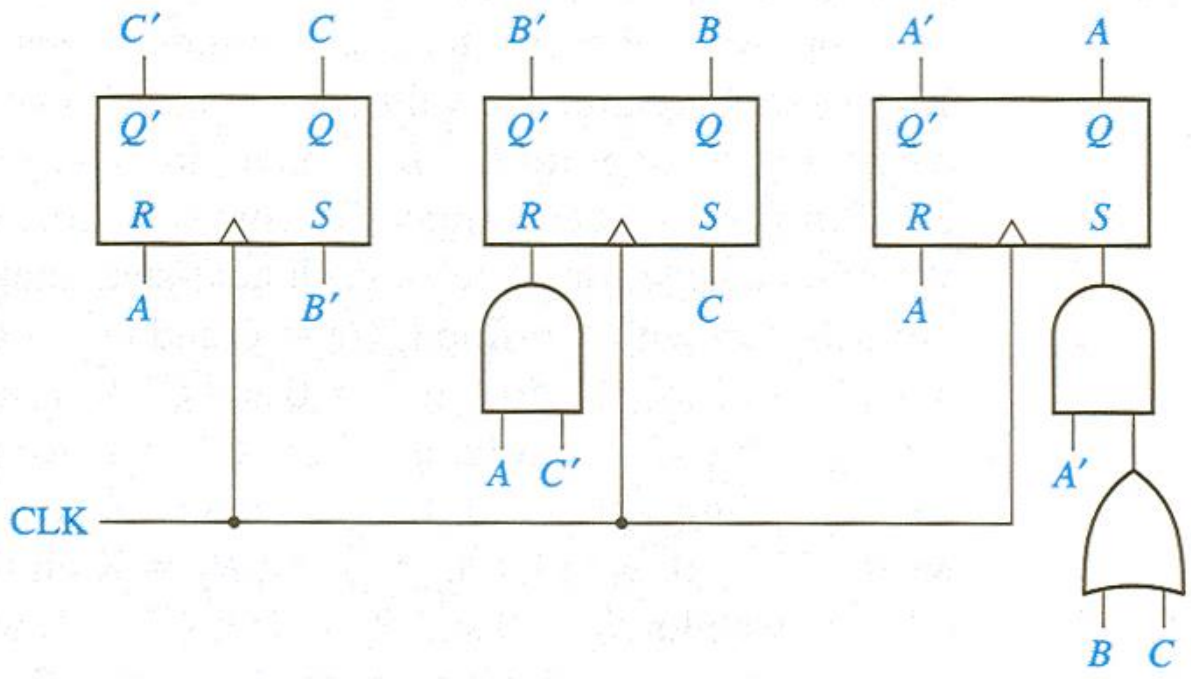
- Logic equation

$R_C = A \qquad S_C = B'$

$R_B = C'A \qquad S_B = C$

$R_A = A \qquad S_A = CA' + BA' = A'(C + B)$

- Counter circuit



Counter Design with J-K FFs (I)

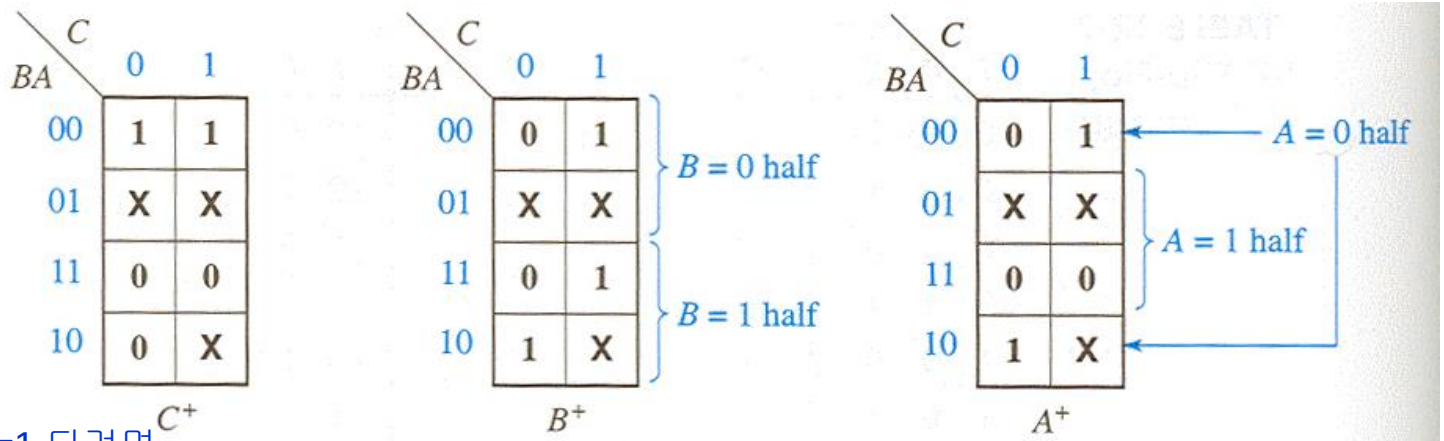
1) Find next state table and map from state graph
2) Find state table for J–K inputs using the next state map
3) Find logic equation of J–K inputs and draw the circuit

Next state table J–K inputs table

C	B	A	C^+	B^+	A^+	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	1	0	0	1	X	0	X	0	X
0	0	1	–	–	–	X	X	X	X	X	X
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	0	0	0	0	X	X	1	X	1
1	0	0	1	1	1	X	0	1	X	1	X
1	0	1	–	–	–	X	X	X	X	X	X
1	1	0	–	–	–	X	X	X	X	X	X
1	1	1	0	1	0	X	1	X	0	X	1

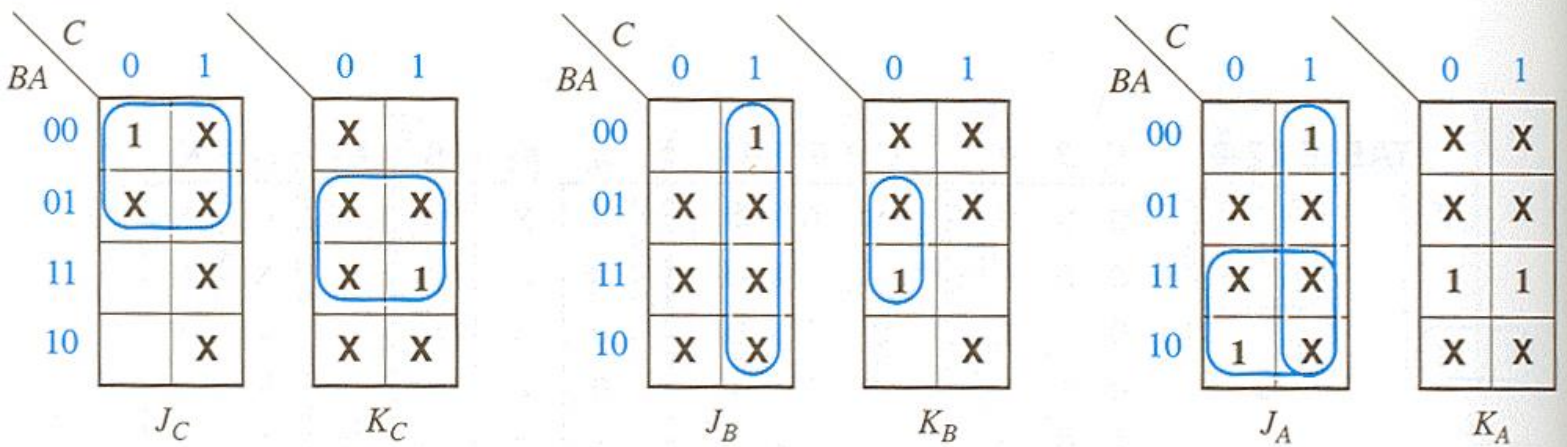
Counter Design with J-K FFs (II)

Find maps of J−K inputs from the next state maps and J−K FF operation



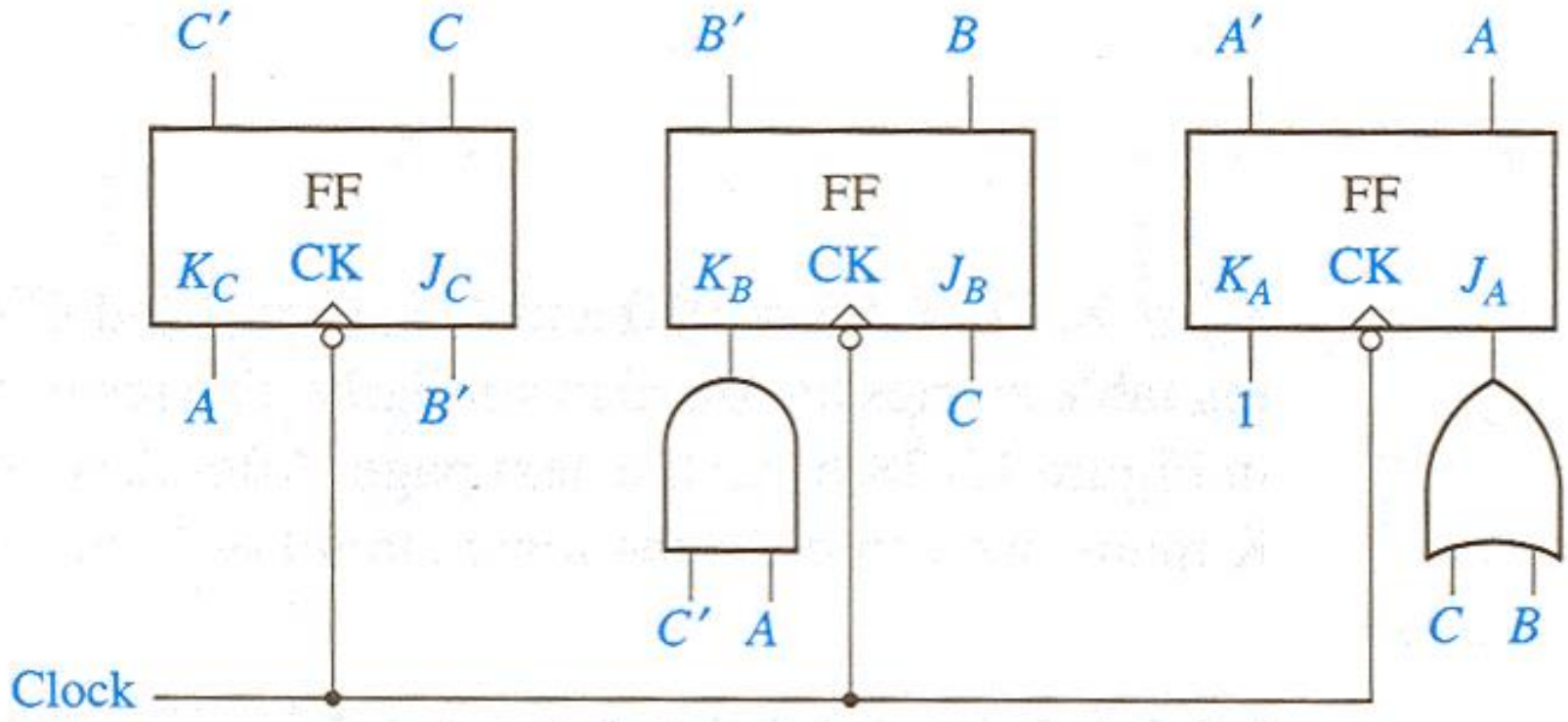
(a) Next-state maps

C=0 일 때, C^+=1 되려면
J_C=1, R_C=x 이어야 한다.



Counter Design with J-K FFs (III)

- Logic equation

$J_C = B'$ $\quad K_C = A$	$J_B = C$ $\quad K_B = C'A$	$J_A = C + B$ $\quad K_A = 1$

- Counter circuit